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FLUID MECHANICS

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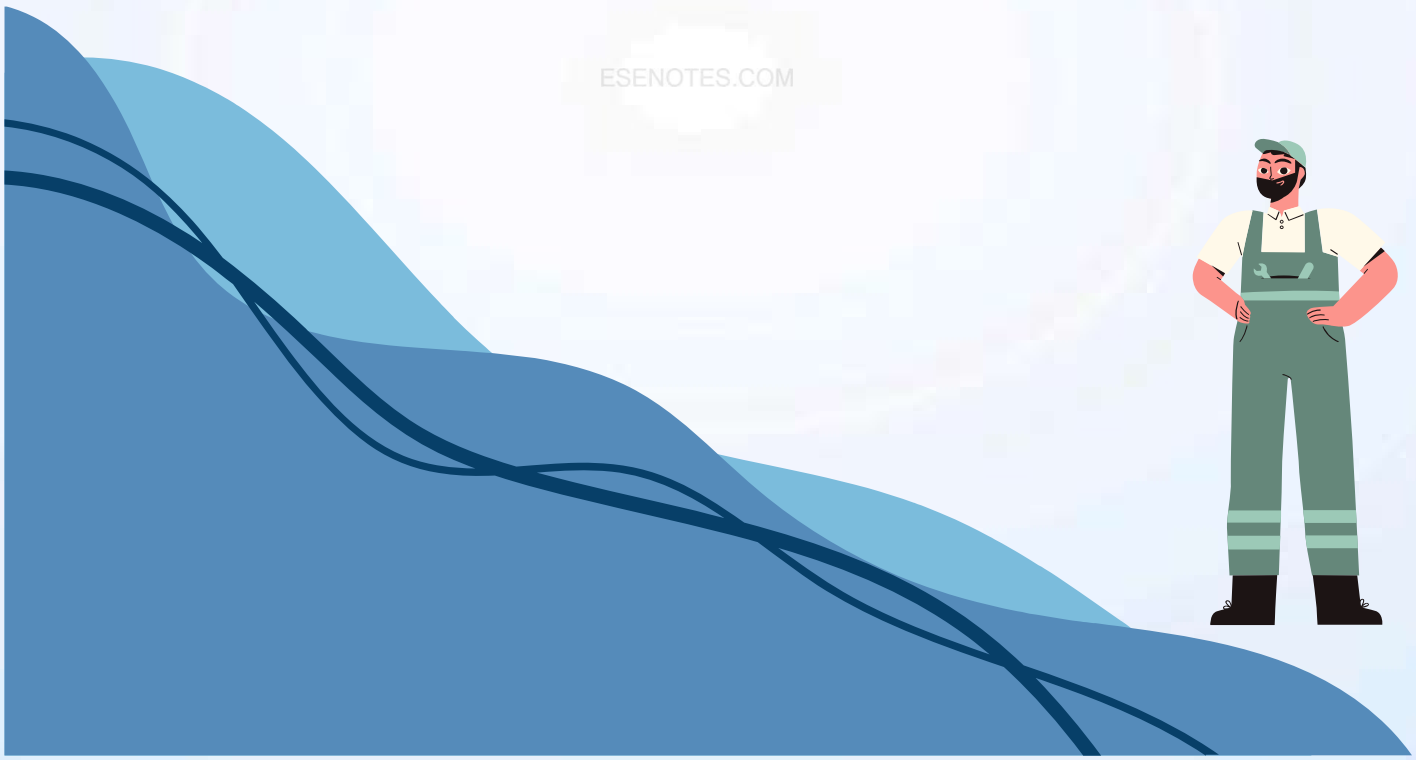

BY JASPAL SIR

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FLUID MECHANICS

- Fluid Mechanics is a branch of engineering, that deals with the properties of the fluid at rest as well as in motion.

- It is further analysed in following forms.

  - Fluid Statics
  - Fluid Kinematics
  - Fluid Dynamics

- Fluid Statics :- It deals with fluid in rest condition.

- Fluid Kinematics :- It deals with fluid in motion. without considering the force responsible for motion.

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- Fluid Dynamics :- It deals with fluid in motion by considering the forces responsible for motion.

- In general matter exist in following states

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(A) SOLID

(B) FLUID → LIQUID, GASES

- A fluid is a substance which exists in LIQUID and GASEOUS phase.

- Fluid is capable of deforming continuously under the action of shear force, shear stress or tangential stress. However small the magnitude of stress may be.

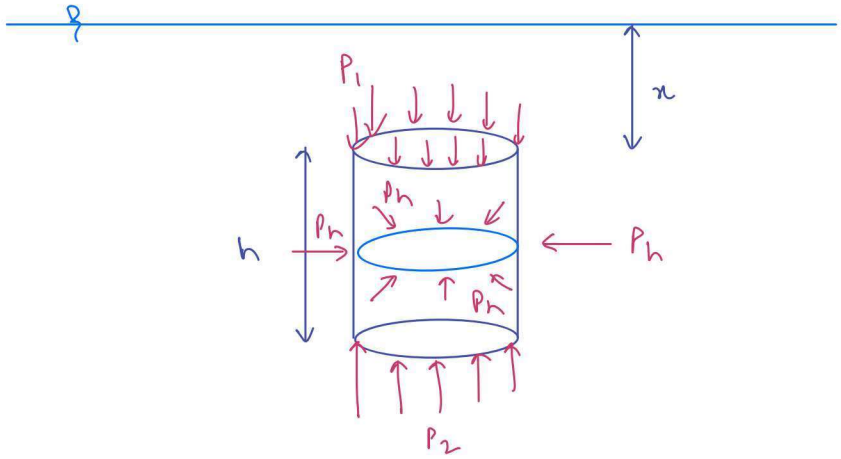
- Hence in solids where stress is propotional to strain, in fluid stress is propotional to rate of deformation or rate of strain.

NOTE :- Here continuous deformation of the fluid means that when a constant shear stress is applied over the fluid, it will continue to deform as long as stress is present.

But in case of SOLID, deformation eventually stops at fixed strain angle.

Buoyancy and floatation

BUOYANCY



$P_1 = \gamma_w x$ $\qquad$ $P_2 = \gamma_w (x+h)$

$F_1 = P_1 \cdot A$ $\qquad$ $F_2 = \gamma_w (x+h) \times A$

Net force in upward direction, $F_{net} = F_2 - F_1$

$= \gamma_w (h+x) A - \gamma_w x A$

$F_{net} = \gamma_w h A$

Here $A \cdot h$ = volume of body = vol. of liquid displaced.

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$\therefore$ $\boxed{F_{net} = \text{wt. of liquid Displaced}}$ ⇒ Archimedes principle.

Buoyant force = Net upward force = wt. of liquid displaced.

July 29

- For a submerged or floating body the resultant force exerted on the body by static fluid is called "buyoant force"
- Buyoant force always act in vertically upward direction and has no horizontal component.
- As per "Archimedes principle", this Buyant force is equal to the weight of liquid displaced.
- point of application of this force at the CG of the displaced liquid and is termed as "CENTRE OF BUOYANCY"

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LAMINAR FLOW

- Laminar flow is a flow in which liquid moves in layers, where one layer slides over another.

- In this case no mixing of different layers take place, hence momentum transfer is not observed. Hence shear force is due to VISCOSITY only.

- In turbulent flow, due to continuous mixing b/w different layers, momentum transfer occurs which gives rise to additional shear.

- In laminar flow, viscous force predominant inertial force.

- For different mediums, nature of flow depends on value of Reynolds number as follows:-

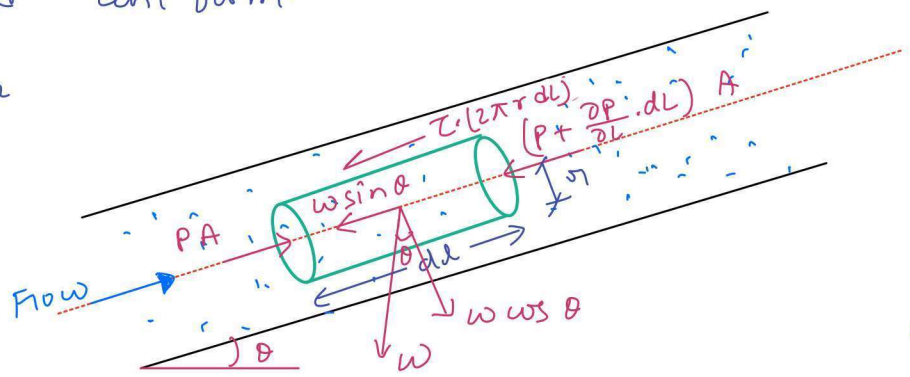
Medium	Nature of flow: Laminar	Transition	Turbulent
1) PIPE	<2000	2000 – 4000	>4000
2) Parallel plate	<1000	1000 – 2000	>2000
3) Open channel	<500	500 – 2000	>2000
4) Soil	<1	1–2	>2

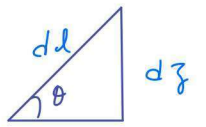
NOTE:- The values of Reynold's no. at which nature of flow changes are not well defined.

FLOW THROUGH CIRCULAR PIPES :-

- In this case flow is always assumed to be steady and uniform.

$A = \pi r^2$



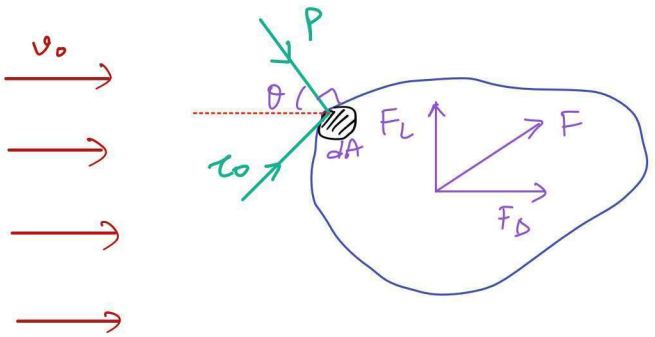


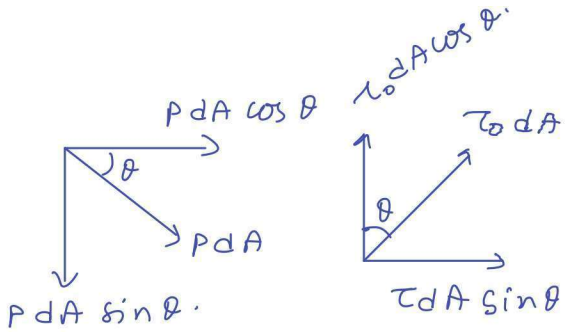
$dl \sin\theta = dz$

$$P(\pi r^2) - P\pi r^2 - \frac{\partial P}{\partial L} dL \cdot \pi r^2 - \gamma \pi r^2 dL \sin\theta - \tau(2\pi r\, dL) = 0.$$

$$-\frac{\partial P}{\partial l} \cdot dL \cdot r - \gamma r (dz) = \tau\, 2\, dL$$

DRAG AND LIFT





- when a body is held in flow of uniform velocity (v_0), the force(F) acting on the body can be resolved in to two components.
  - The force acting in the direction of motion is termed as Drag force.
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  - And the force acting ⊥r to the direction of flow is termed as LIFT FORCE (F_L).
  - Since the velocity at different at different points near the surface of body are different, application of pressure at this points also varies.
  - At any point on the surface of the body, two types of forces are acting.

(a) Shear Stress (τ_0) :- Acting in tangential dirn.

(b) Pressure force (P), Acting in Normal direction to the surface.

- Hence drag force acting in direction of motion is given by the summation of the components of shear force and pressure force acting over the entire surface of body

Drag force, $$F_D = \int_A \tau_0 dA \sin\theta + \int_A P dA \cos\theta.$$

- And lift force acting in direction of flow is given by.

$$F_L = \int_A \tau_0 dA \cos\theta + \int_A P dA \sin\theta.$$